ITU-MiniTwit - Group e - Report

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4.1 Evolution and refactoring

Authors: Frederik

For the simulator to work there needed to be added some new endpoints which were specified in the swagger documentation. To make the work easier the OpenAPI Generator CLI tool was used and the different endpoints specialized in what was expected of them. Furthermore, authorization using a basic token was implemented to set up for the different endpoints that need it.

The database for the application was a SQLite database running in the minitwit application. That meant that the persisted data the database was holding got deleted when a new deploy of the application occurred. There was therefore a need to migrate to a new database type so the data is persisted. It was chosen to migrate to a MySQL database which was hosted using Digital Ocean. This was chosen for its simplicity and time effectiveness for the group.

4.2 Operations

Authors: Frederik

Checking monetoring and logs often to see if there is exes strain on the system and if any errors have been thrown. Furthermore check of the status page to see if any errors has been thrown by the simulator that the loging did not catch.

4.3 Maintenance

Authors: Fredrik

api errors

We were made aware that Grafana was returning an internal server error when trying to log in. It was discovered that Prometheus had filled the droplets storage so it could not try to access the internal volume to log in. To fix this a folder docker uses to write data to was deleted and the docker containers on the VM were restarted. To make sure this did not happen again storage retention was added Prometheus got its own volume. To see the whole bug report see issue [#81(https://github.com/Itu-DevOps-2026/ITU-MiniTwit/issues/81)].

indexing of cheeps

4.4 Reflect and describe what was the "DevOps" style of work

5 Use of Generative AI

Authors: Nikolej

During the development of this project we used ChatGPT in two main ways:

Debugging. Often when encountering unexpected bugs and error messages, we would consult ChatGPT about the cause and potential fixes While it could rarely fix the issues entirely by itself, more often than not, it pointed us in the right direction.

Research. This course presents challenges involving numerous technologies, most of which we were unfamiliar with. As such, ChatGPT was used to summarize lengthy documentation and provide concrete guides tailored to our situation.

Generating boilerplate / configurations. ChatGPT was also used for simple tasks like generating boilerplate code or writing Github Actions Workflow files. For example, the `build-report.yml` workflow, that converts the markdown source into a pdf.

The use of AI has made these tasks easier, spared us many frustrations and saved consiberable time during development. On the flip side, AI may have deprived us the extensive knowledge and experience you gain by, for example, painstaking reading of documentation or spending hours resolving a tiny bug.